Mason Bane

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Summary: Computer Science student (Computer Engineering) targeting embedded firmware and hardware/software integration roles. Proficient in microcontroller programming, low-level C/C++ and ARM Assembly, and circuit design. Strong mathematical/analytical skills with research experience.

Education

Tarleton State University — Stephenville, TX
Bachelor of Science in Computer Science, Concentration in Computer Engineering — *Aug. 2022 – May 2026 (Expected)*
Minor in Mathematics
GPA: 3.94/4.00 (Institutional) — Cumulative: 3.75/4.00

Hill College — Hillsboro, TX
Associate of Arts in Liberal Arts — *Aug. 2019 – Sept. 2023*

Experience

Undergraduate Research Assistant - Lead Programmer — May 2024 – Present
Tarleton State University — *Stephenville, TX*

- Developed and optimized of various N-Body simulations utilizing C/C++, enhancing computational efficiency and accuracy through advanced algorithms and parallel processing with CUDA.
- Collaborated with interdisciplinary teams to create digital twins to model complex problems using OpenGL and Blender, translating scientific concepts into interactive simulations and improving data interpretation.
- Conducted rigorous testing and debugging of simulation software, ensuring robust performance and reliability while documenting processes to facilitate knowledge transfer and future research initiatives.

Undergraduate Technology Specialist - HPC Lab Manager — Aug. 2024 – Present
Tarleton State University — *Stephenville, TX*

- Managed and maintained 15 Linux devices in a high-performance computer lab dedicated to research initiatives
- Performed system updates, hardware maintenance, and technical support, resolving issues promptly to minimize downtime and maintain optimal performance.
- Maintained a clean and organized lab environment, promoting a collaborative workspace that fosters innovation and productivity.

Projects

Secure IoT Sensor Node with Hardware TrustZone | *C, STM32CubeIDE/HAL, I2C/SPI* — Dec. 2025 – Present

- Implementing a secure data pipeline on an STM32H5 MCU, utilizing Arm TrustZone and the Secure Manager to create a hardware-rooted chain of trust for environmental sensor data.
- Developing drivers and application logic to interface with a Bosch BME280 temperature, humidity, and pressure sensor via I2C communication protocol.
- Structuring firmware to cryptographically sign sensor readings within the secure processing environment for tamper-evident logging.
- Utilizing STM32CubeIDE and the HAL for peripheral configuration and hardware security module (HSM) management as a foundation for secure telemetry systems.

N-body Digital Twin of the Left Atrium | *NIH Grant #1R15HL179671-01* | *C, CUDA* — Aug. 2024 – Present

- Engineered a parallel N-body model of the left atrium using CUDA, with a 20,000+ node mesh to simulate atrial arrhythmias in near real-time.
- Developed an intuitive C++/ImGui interface to control simulation parameters and visualize outputs, bridging the gap between complex computational models and end-user (research/clinical) requirements.
- Presented findings at academic conferences, demonstrating the tool's potential to improve clinical decision-making and transform training for medical professionals.

Technical Skills

Programming Languages: C, C++, Python, ARM Assembly (Cortex-M), Java, Bash
Microcontroller Platforms: STM32 (H5/Cortex-M33), TIVA-C (TM4C), Arduino, Raspberry Pi
Embedded Protocols & Tools: I^2C, SPI, UART, GPIO, STM32CubeIDE, GDB, RTOS Concepts
Systems & Development: Git/GitHub, Linux, CMake/Make, Visual Studio, VS Code
Simulation & Design: LTSpice, MATLAB, CUDA, OpenGL, ImGui, Blender